FastF1 Sim

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1 Abstract

Using the FastF1 Python library to create a quasi-steady state Formula 1 lap time simulator, simulating a qualifying lap around the Silverstone circuit, using estimated vehicle performance parameters from available databases producing a lap that is marginally faster than Max Verstappen's 2025 pole position lap.

2 Introduction

Simulating vehicle performance is arguably the cornerstone of modern motorsport, enabling the evalation of vehicle setup and performance at a fraction of the resources required to physically test. By modelling the interaction between a car's aerodynamics, performance, and powertrain, these simulations translate engineering parameters into the ultimate performance metric - lap time.

There are various methods to model transient multi-body dynamics into a simplified point-mass approach. This project utilises a quasi-steady-state(QSS) approach, wherein the vehicle is treated as a point mass, with the assumption it is in a state of instantaneous force balance at every point on the track. This neglects yaw dynamics and suspension responses, however captures the prominent physics governing lap time; the balance between grip availability, demands of cornering, braking and acceleration, at a minimal computational cost, enabling a thorough first principles study.

This project aims to develop a forward QSS lap time simulator, parametrised from estimated vehicle characteristics, validating accuracy against 2025 British Grand Prix telemetry. Physical parameters are chosen prior to simulation, and the resulting prediction is compared to the pole lap as an independent benchmark.

We first look into the method behind each component of the simulator, track reconstruction, tyre modelling, and the solver upon which the entire simulation relies, before validating results and evaluating critical limitations of this model.

3 Method

The simulator model comprises of four stages, circuit reconstruction, modelling forces available at the tyre contact patch, breaking down forces on the car, and a QSS solver which implements these to produce a speed profile and lap time. Each stage is described below.

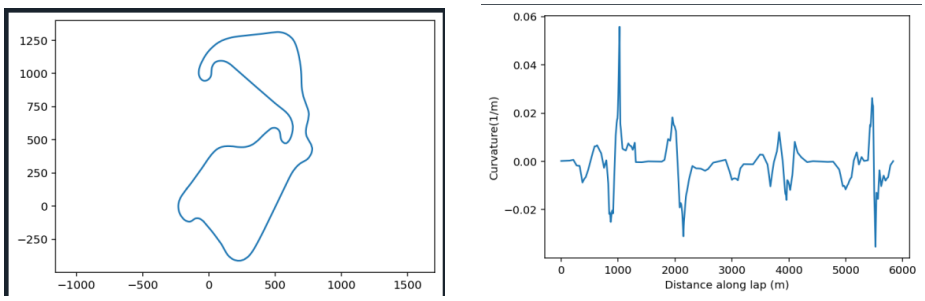
3.1 Track Reconstruction

In order to simulate lap time, the circuit geometry is expressed as curvature κ as a function of the distance travelled, s. The track was constructed using positional data from a single flying lap, obtained through the FastF1 library. The car's recorded (x, y) coordinates as it traced its path around a given track. However it is unsuitable to numerically differentiate these discrete points, due to second derivatives amplifying noise in measurement.

In order to accurately map the circuit, a periodic cubic spline was aligned with the coordinate data, providing a smooth representation of the path through which second derivates could be computed. The curvature at each point is computed as

$$\kappa = \frac{x'y'' - x''y'}{(x'^2 + y'^2)^{3/2}} \tag{1}$$

where the prime symbol represents differentiation with respect to the spline parameter. In order to find the arc length, the path speed $\sqrt{x'^2 + y'^2}$, resulting in $\kappa(s)$ being sampled over a uniform distance grid, facilitating the numerical solver. The reconstructed circuit and curvature profile are shown in Figures 1(a) and 1(b) respectively.



(a) Reconstructed circuit path.

(b) Curvature profile $\kappa(s)$.

Figure 1: Track reconstruction from positional telemetry.

3.2 Tyre Model

The tyre behaviour was characterised using the Pacejka Magic Formula, which describes the relationship between tyre slip angle and lateral force. The peak

lateral force is load-dependent, with the effective friction coefficient decreasing as vertical load increases. This load sensitivity was represented using a simplified empirical relationship.

For the lap-time solver, the available lateral force was approximated by the peak grip, $F_{y,\max} = \mu(F_z)F_z$, rather than solving for tyre slip angle throughout the lap. This simplification enables the maximum cornering speed to be calculated efficiently, although it does not capture the full force response of the tyre under varying slip conditions.

$$F = D\sin\left(C\arctan\left(Bx - E\left(Bx - \arctan(Bx)\right)\right)\right) \tag{2}$$

where x is the slip angle, the coefficients B, C, D, and E dictate the stiffness, shape, peak and curvature of the force characteristic respectively. Peak force is load dependent, with $D = \mu F_z$. However, the coefficient of friction μ decreases with increasing vertical load, this phenomena is called load sensitivity, modelled as

$$\mu(F_z) = \mu_0 - k(F_z - F_{z0}) \tag{3}$$

where F_{z0} and μ_0 are the friction and force values for a known reference point. For a vehicle with high downforce, this effect is significant, as the aerodynamic loads at high speed yield less than a proportional grip increase. Figure 2 shows the characteristic force curves produced by this model.

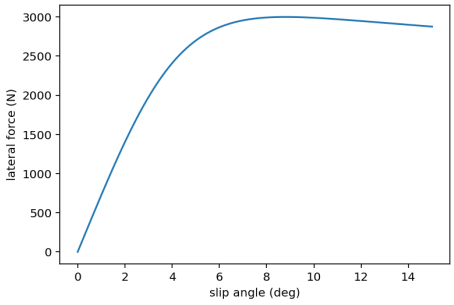


Figure 2: Characteristic force curve produced through this model

3.3 Aerodynamic and Vehicle Model

The vehicle was modelled as a point mass, and aerodynamic forces were assumed to scale with the velocity squared, according to $F = \frac{1}{2}\rho v^2 CA$, characterised by the lift and drag coefficients C_L and C_D. The vertical load F_z is equal to the sum of the weight of the vehicle and the aerodynamic downforce, thus $F_z = mg + \frac{1}{2}\rho v^2 C_L A$. The parameters used were estimated from publicly available data, not fitted to a reference lap initially. Table 1 shows the vehicle and aerodynamic parameters used to run the simulation.

Table 1: Vehicle and aerodynamic parameters used in the simulation.

Parameter	Symbol	Value	Source / justification
Vehicle mass	m	798 kg	2025 FIA minimum weight
Effective power	P	680 kW	Qualifying-averaged estimate
Downforce coefficient	$C_L A$	4.0	Representative high-downforce value
Drag coefficient	$C_D A$	1.1	Representative F1 value
Peak friction	μ_0	1.7	Typical dry slick
Braking deceleration	a_{brake}	$45\,\text{m/s}^2$	$\approx 4.6\,g$

3.4 Quasi-Steady-State Solver

At each point on the circuit, it is assumed that the car is in a state of instantaneous force balance. The maximum cornering speed is set by the condition that the lateral grip must supply the required centripetal force.

$$mv^2\kappa = \mu\left(mg + \frac{1}{2}\rho v^2 C_L A\right) \tag{4}$$

μ depends on the load, hence on the speed. This was solved iteratively at each point. However, the resulting cornering-limited speed does not account for the finite rates at which the vehicle can brake and accelerate. Thus, two sequential passes were applied: a backward pass limiting the speed achievable when braking into a corner, and a forward pass limiting the acceleration out of a corner, with the instantaneous speed being the minimum of the two. The start and finish conditions are continuous and the passes applied periodically, as the circuit is a closed loop. The lap time was obtained by integrating the inverse of the speed over distance.

$$t = \int \frac{ds}{v(s)} \tag{5}$$

4 Results

The result was run using the parameters highlighted in the table 1, and the output was compared against M. Verstappen's qualifying lap. The below table shows the simulated speed profile as a function of distance around the lap.

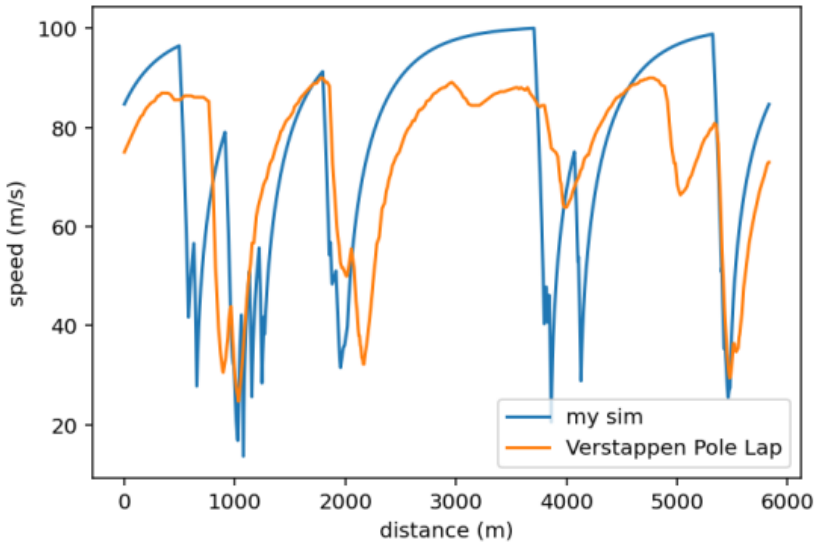


Figure 3: Simulated speed profile compared with the measured pole lap of the 2025 British Grand Prix.

The simulator predicted a lap time of 84.295 s, compared with the recorded pole time of 84.892 s, the simulator being about six tenths of a second faster over the lap. The comparison reveals three distinct regions of behaviour. Through the high-speed corners the predicted and measured speeds agree closely. On the straights the model slightly over-predicts the top speed, reaching approximately 100 m/s against a measured value near 90 m/s. Conversely, at corner entry and through the slowest corners the simulated speed is consistently lower than measured, with the model braking earlier and reaching a lower minimum speed.

The close agreement in overall lap time therefore arises in part from these opposing errors acting in different sections of the lap. Their origins are examined in Section 5.

5 Discussion

The agreement in overall lap time, achieved using independently-estimated parameters rather than values fitted to the reference lap, indicates that the model captures the dominant physics governing performance around the circuit. The residual discrepancies identified in Section 4 are instructive, as each can be attributed to a specific simplification in the model.

The most significant deviation occurs at corner entry, where the simulated car consistently brakes earlier and carries less speed than the real vehicle. This follows directly from the model treating braking and cornering as sequential rather than simultaneous. In reality a driver trail-brakes, continuing to brake while turning in and exploiting the tyre's combined longitudinal and lateral

grip. The quasi-steady-state solver requires the vehicle to have reached the pure cornering speed before turn-in, and so necessarily underestimates entry speed.

A second deviation appears on the straights, where the model over-predicts the top speed. This reflects the representation of the power unit as a single constant power. The real car's deployment varies around the lap, and the constant-power assumption, combined with the neglect of secondary losses such as drivetrain friction, yields a slightly optimistic straight-line speed.

A further simplification is the fixed racing line. The model is supplied with the recorded path of a single lap and cannot optimise the line to trade cornering radius against speed, as a real driver continuously does. The point-mass representation similarly neglects load transfer and any distinction between front and rear axle grip.

These limitations act predominantly to reduce cornering speed, while the constant-power assumption inflates straight-line speed; the close agreement in integrated lap time therefore reflects a degree of cancellation between errors of opposing sign. The result should accordingly be interpreted as validation of the model's overall fidelity rather than of its local accuracy at every point on the circuit.

6 Conclusion

A quasi-steady-state lap time simulator was developed and validated against real Formula 1 telemetry. Circuit geometry was reconstructed from positional data, tyre behaviour modelled through a load-sensitive Pacejka formulation, and the speed profile obtained using a periodic forward–backward solver. Using physically-motivated parameters, the model predicted the 2025 British Grand Prix pole lap time to within 0.7%, with the residual error traceable to identifiable structural simplifications and potentially parameter uncertainty.

Several extensions would address the principal limitations identified above. A combined-slip tyre model incorporating the friction ellipse would capture trail-braking and is expected to improve corner-entry fidelity, while a grip-limited traction model would improve the prediction of corner exits. Extending the simulation to further circuits would test its generality, and parameter inference from telemetry could be used to characterise the effective properties of a real vehicle. Collectively these would develop the model from a validated single-lap predictor toward a tool capable of exploring setup and configuration trade-offs.